

CashFlow Sentinel

Explainable Cash-Flow Early-Warning System for Indian MSME Suppliers

Exathon Hackathon | Final Submission

THE PROBLEM

Late & Inconsistent B2B Payments Cripple Indian MSMEs

₹10+ Lakh Crore

Outstanding receivables across Indian MSMEs

Manual Reconciliation

Spreadsheet-based matching of invoices & payments

Missed Discrepancies

Duplicate invoices, partial payments go undetected

No Prioritisation

Suppliers don't know which invoices to chase first

TARGET USER

Indian MSME Suppliers Managing Receivables

- ✓ Small and medium suppliers managing 50–500+ invoices per month
- ✓ Dealing with partial payments, deduction disputes, and delayed remittances
- ✓ Using Tally, spreadsheets, or manual ledgers for reconciliation
- ✓ Losing working capital to undetected mismatches and unprioritised collections
- ✓ Need actionable intelligence — not just data entry

OUR SOLUTION

A Dashboard That Ranks Collection Work & Highlights Evidence-Backed Discrepancies

Risk-Scored Invoices

Transparent 0–100 heuristic score with 4 explainable factors

Auto Reconciliation

Detects duplicates, partial payments, mismatches & unmatched remittances

Collection Priority

Ranked action queue — tells you which invoice to chase first

Audit Trail

Every reconciliation run logged in Exasol with full traceability

ARCHITECTURE

Full-Stack, Reproducible, Enterprise-Grade

FRONTEND

React + Vite

- Interactive dashboard UI
- 4 tabs: Overview, Invoices, Discrepancies, Collections
- Real-time API calls
- ₹ INR formatting
- Responsive design

localhost:5173

BACKEND API

Flask + Python

- RESTful /api endpoints
- Risk scoring engine
- Reconciliation rules
- Deterministic analytics
- CORS-secured
- Stateless query layer

localhost:5000

DATABASE

Exasol Personal

- In-memory columnar OLAP
- CFS_DEMO schema
- Version-controlled DDL
- MERGE-based seeding
- Audit event logging
- Enterprise-grade

127.0.0.1:8563

EXPLAINABLE RISK ENGINE

Transparent 0–100 Heuristic — Not a Black Box

Every score is traceable. The supplier sees exactly which factor contributed how many points.

Historical Late Days — 35 pts

avg_days_late × 3.5

Invoice Timing — 30 pts

Overdue days × 2.5 or proximity to due date

Open Balance Size — 20 pts

Outstanding balance ÷ ₹5,000

Late-Payment Rate — 15 pts

Customer historical late-payment % × 15

Score ≥ 70 → HIGH RISK

Score ≥ 40 → MEDIUM RISK

Score < 40 → LOW RISK

RECONCILIATION ENGINE

5 Deterministic Rules — Evidence, Not Guesswork

HIGH	**Duplicate Invoice**	Same invoice number appears more than once e.g., AST-1048 occurs 2×
HIGH	**Invoice/Payment Mismatch**	Invoice marked 'paid' but payment total ≠ invoice value e.g., BRF-2060: ₹91,000 invoice, ₹85,000 payment
MEDIUM	**Partial Payment**	Invoice marked 'partial' has remaining balance e.g., BRF-2051: ₹68,000 invoice, ₹40,000 paid
MEDIUM	**Unmatched Payment**	Payment received with no linked invoice e.g., UNMATCHED-778: ₹15,000 with no invoice ID
MEDIUM	**Deduction Reference**	Payment reference contains 'DEDUCTION' — needs evidence e.g., DEDUCTION-UNEXPLAINED on ₹5,000 payment

DATA MODEL

4 Entities · CFS_DEMO Schema · Exasol DDL

CUSTOMERS

id, name, avg_days_late, late_rate

Historical payment behaviour per buyer

INVOICES

id, invoice_number, due_date, amount, status, customer_id

Receivable registry with status tracking

PAYMENTS

id, invoice_id, amount, payment_date, reference

Settlement evidence and reconciliation matching

AUDIT_EVENTS

id, event_type, created_at, details

Reconciliation run traceability log

REST API

7 Clean Endpoints — Flask-Powered

METHOD	ENDPOINT	DESCRIPTION
GET	/api/health	Active store verification (exasol/sqlite)
GET	/api/dashboard	KPI metrics, expected collections, top alerts
GET	/api/invoices	Paginated, filterable by query, status, risk level
GET	/api/invoices/<id>	Full invoice detail + payments + collection priority
GET	/api/alerts	Suspected discrepancies, filterable by severity
GET	/api/collections	Ranked collection action queue
POST	/api/reconciliation/run	Run reconciliation, log audit event, return count

WHY WE STAND OUT

Honest, Traceable, Production-Quality Thinking

✗ Others: Black-box AI claims	✓ **Us: Explainable heuristic — every point traceable**
✗ Others: SQLite / mock data	✓ **Us: Exasol in-memory columnar OLAP database**
✗ Others: Figma mockups	✓ **Us: Live working full-stack prototype**
✗ Others: Generic global problem	✓ **Us: India-specific MSME receivables focus**
✗ Others: No testing	✓ **Us: Unit, API, and integration tests + CI-ready**
✗ Others: No documentation	✓ **Us: Architecture docs, pitch deck, security policy**

TESTING & QUALITY

Production-Grade Rigour — Not Just a Demo

Unit Tests

Risk scoring validation with known inputs

- High risk for late + large balance
- Low risk for recent + small balance

API Tests

All endpoints tested including:

- Dashboard data derivation
- Alert type coverage (duplicates, unmatched)
- Input validation (400 for bad params)
- Reconciliation success + failure paths

Exasol Integration

End-to-end test: seed → query → verify

- Row count assertions on live Exasol
- Reconciliation audit trail verification
- Graceful skip when credentials absent

Code Quality

- Version-controlled DDL (exasol_schema.sql)
- Idempotent MERGE-based seeding
- .gitignore covers .env, SQLite, caches
- SECURITY.md + CONTRIBUTING.md

IMPACT & HONEST LIMITATIONS

What It Does — And What It Doesn't

✓ IMPACT HYPOTHESIS

- → Faster prioritisation of risky invoices
- → Systematic detection of payment anomalies
- → Audit-logged reconciliation for compliance
- → Reduced manual spreadsheet matching effort
- → Transparent scoring builds supplier trust
- → Architecture ready for real-world data plug-in

⚠ HONEST LIMITATIONS

- → Synthetic demonstration data only
- → Heuristic scoring — not a trained ML model
- → Human investigation always required
- → Does not prevent fraud or guarantee payment
- → Requires real-world validation before claiming savings
- → Local-only prototype — not cloud-deployed

FUTURE ROADMAP

From Prototype to Production — The Path Forward

1

PHASE 1

Data Integration

Connect to Tally exports, GST e-invoice feeds, UPI remittance data

2

PHASE 2

Scale & Deploy

Cloud deployment on Exasol SaaS, multi-tenant support, role-based access control

3

PHASE 3

Intelligent Scoring

ML model trained on real payment histories, validated against current heuristic

4

PHASE 4

MSME Ecosystem

Supplier network scoring, credit risk feeds, automated follow-up workflows



Thank You

CashFlow Sentinel — Explainable Cash-Flow Intelligence for Indian MSMEs

7	4	5	3	100%
API Endpoints	Risk Factors	Recon Rules	Test Suites	Reproducible